



Banco — Backup & Disaster Recovery

How the pieces fit, and exactly how to rebuild from a laptop + a password

LA PIAZZA · BANCO

lose the shop, keep the data

THE PIECES (7 BLOCKS)

 **The Box** — the Hetzner server. Runs the till AND makes the backups. Always on.
46.62.138.218

 **The Backup** — a full copy of the database, made **every night at 3am** by itself.
banco_prod_YYYYMMDD_HHMM.sql.gz.gpg

 **The Lock** — GPG AES-256. Scrambles each backup so the file is useless without the passphrase.
gpg --symmetric

 **The Vault (B2)** — Backblaze off-site storage. Copies here **can't be deleted for 14 days** (ransomware-proof).
bucket: ecolution-banco-backups

 **The Read-Key** — your Backblaze login that pulls a copy **out** of the vault. Read-only.
B2 app key · in KeePass

 **The Passphrase** — the GPG password that **unscrambles** a backup back into a database.
64 chars · in KeePass

 **KeePass** — the safe on your laptop that holds **both keys**. One master password (in your head) opens it. This is the single thing you must protect.
KeePass-Helix-DB.kdbx · also synced to Google Drive

HOW THEY SNAP TOGETHER — EVERY NIGHT, 3AM, NO ONE TOUCHES IT

- 1 Copy** — the box dumps the whole banco_prod database.
- 2 Lock** — GPG scrambles it (AES-256). Now it's an unreadable .gpg file.
- 3 Prove** — the box *unlocks its own copy and reloads it* into a throwaway database, and checks the row counts match. A backup that can't restore is caught here, not in a disaster.
- 4 Ship** — the locked copy goes to the B2 vault, stamped immutable for 14 days.
- 5 Signal** — the box pings the alarm: "I'm alive, backup done." **If any step fails — or the box goes dark — no ping arrives, and the alarm emails you** within 2 hours.

TWO LOCKS, TWO KEYS — THE ONE THING TO REMEMBER

 **The Read-Key** opens the **VAULT**
gets the sealed file OUT of Backblaze B2. Can't read what's inside it.

 **The Passphrase** opens the **FILE**
unscrambles the sealed file into the real database. Can't get it out of B2.

Neither key alone is enough — steal one, you get nothing. Both live in KeePass, behind your master password. That's the whole security model in one sentence.

```
# 0 — open KeePass (master password) → copy out the Read-Key (id+key) and the GPG
passphrase

# 1 — log into the B2 vault with the Read-Key
b2 account authorize «read-keyID» «read-appKey»

# 2 — find the newest backup, then pull it out of the vault
b2 ls b2://ecolution-banco-backups/banco/
b2 file download b2://ecolution-banco-backups/banco/«newest».sql.gz.gpg
/tmp/restore.gpg

# 3 — unscramble it with the GPG passphrase (paste it when asked)
gpg --pinentry-mode loopback -d /tmp/restore.gpg | gunzip > /tmp/restore.sql

# 4 — load it into a fresh Postgres → the database is back
createdb banco_prod && psql -d banco_prod < /tmp/restore.sql
```

✓ Proven end-to-end on 2026-07-19: steps 1–3 pull today's immutable copy from B2 and decrypt it to -- PostgreSQL database dump, with the box completely out of the picture. That's not a claim — it's a drill you ran.

WHAT THIS GUARANTEES YOU

- ✓ **Three places** — the box, the B2 vault, and off-box copies. Lose any one, lose nothing.
- ✓ **Immutable 14 days** — a hacked box can add copies but can't erase the ones already in the vault.
- ✓ **Self-verified nightly** — every backup is test-restored, so it's *proven*, not hoped.
- ✓ **Alarm on silence** — a failed or missed backup emails you within 2 hours.

STRAIGHT TALK — SCOPE

This recovers the **data** (the database — the thing that's irreplaceable), and it's proven. Rebuilding the whole *server* (a fresh box, Docker, Keycloak) is a separate, longer job — hours following the ops runbook — because Banco runs on one server today. The keys live in KeePass; if you ever lose the KeePass master password, nothing above can help — that password is the one irreplaceable thing.